Xinyu Shen

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Portfolio: https://yuki-portfolio-three.vercel.app/
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Summary

Senior at the University of Florida double majoring in Data Science and Economics. Have hands-on experience applying machine learning, statistical programming, and health/economic data analysis to research and analytics projects. Interested in graduate study and roles connecting statistical modeling, healthcare analytics, and economic decision making.

Education

University of Florida, Gainesville, FL B.S. in Data Science & Economics | GPA: 3.85 | Expected May 2027

Internship/Research Experiences

Student Researcher, Dr. Brant Research Group at University of Florida May 2026 to August 2026

- Conducted simulation studies for the SMFsim R package to evaluate how alpha-level tuning affects model performance in within-group and between-group statistical comparisons.Organize and document simulation settings, compare model outputs across multiple parameter combinations, and assist with reproducibility, interpretation, and visualization of statistical results.
- Tested multiple alpha values across simulation settings and found that adjusted alpha thresholds improved performance by normalizing small effects while preserving larger signal effects.
- Compared the SMFsim approach against raw data, ComBatMat-adjusted data, and downsampled data to evaluate robustness across alternative preprocessing and correction strategies.
- Troubleshot data-filtering and simulation workflow issues, including adapting the analysis pipeline after the initial dataset produced too few eligible observations under project-specific filtering criteria.
- Ran and organized large-scale simulation workflows using the University of Florida HiPerGator supercomputing environment, documenting parameter settings, model outputs, and performance comparisons for reproducible analysis.

Biostatistics & Data Science research internship, Florida Atlantic University June 2025 to August 2025

- Selected for an NIH-funded training program in applied biostatistics, statistical learning, and data science using R, including tidyverse based data wrangling and visualization.
- Collaborated on an interactive R Shiny app to predict stroke risk using multiple binary classification models and tidyverse workflows for preprocessing and analysis.
- Cleaned and preprocessed clinical stroke data for modeling using R packages such as tidyverse, dplyr, and ggplot2, applying dummy/one-hot encoding, missing-value imputation, and SMOTE oversampling to address class imbalance and improve training data quality.
- Built, tuned, and compared classification models using R-based workflows, including logistic regression, LASSO, and tree-based approaches such as random forest, validating results with confusion matrix, accuracy, precision/recall, and F1-score when applicable.
- Developed an interactive R Shiny application by designing pages, tabsets, and reactive components so users can explore visualizations and compare model performance across original vs SMOTE-resampled datasets.
- Supported faculty-led research by managing datasets and running statistical analyses in SAS and SPSS, including importing and merging files, cleaning data, recoding/creating variables, producing descriptive statistics (frequencies, means, cross-tabs), and generating regression outputs and summary tables for reporting and presentations.

Projects

AI Adoption and Hospital Efficiency, Empirical Research in Economics, University of Florida. Fall 2026(Planned)

- Proposed an empirical study examining whether AI-related health IT adoption improves hospital cost efficiency and productivity from an economic perspective.
- Plan to combine CMS Hospital Provider Cost Report data with ONC/HealthIT.gov hospital technology adoption measures to build a hospital-level dataset.

- Expected outcomes include cost per discharge, cost per patient day, discharges per FTE, and possible stochastic frontier analysis or DEA-based efficiency scores.

DataFest 2025, University of Florida March 2025

- Collaborated on a rapid-turnaround analytics project, cleaning, merging, and validating large housing datasets using JMP, R, and Python; handled missing values, standardized fields, and engineered derived variables tied to pricing and demand.
- Conducted exploratory data analysis (EDA) and statistical modeling (regression, correlation, and segmented comparisons) to identify key predictors and quantify relationships driving housing outcomes.
- Built clear visualizations and reporting outputs (plots, dashboards-style summaries, and formatted tables) to communicate patterns in affordability, location effects, and demographic differences across groups.
- Delivered a concise, data-driven presentation for judges, translating technical findings into actionable recommendations while clearly stating assumptions, limitations, and potential sources of bias.

Minesweeper Project with C++, University of Florida December 2024

- Designed and implemented a complete Minesweeper game in C++, featuring a welcome/menu screen, interactive gameboard, and a persistent leaderboard to track and display player results across runs.
- Applied object-oriented programming (OOP) and modular architecture to separate core game logic (board initialization, mine placement, adjacency counting, reveal/flag actions, win/loss detection) from UI rendering and event-driven input handling.
- Integrated professor-provided graphics assets by loading image files via file paths and mapping sprites/icons to tiles and game states, enabling consistent rendering for reveals, flags, and win/loss screens.
- Implemented robust gameplay state management including a timer, mine counter, pause/resume controls, and a dynamic face icon that updates based on game state (in-progress, win, loss) for a polished user experience.

Cardiovascular Disease Prediction Machine Learning Model, University of Florida April 2024

- Built and compared supervised classification models (Logistic Regression, Decision Tree, Random Forest, and SVM) on a cardiovascular dataset (~70K records) to predict disease risk and support early-risk screening use cases.
- Cleaned and preprocessed clinical features (e.g., age, blood pressure, cholesterol, glucose), performed train/test splits, and evaluated model performance using accuracy and other standard classification metrics (e.g., confusion matrix, precision/recall as applicable).
- Tuned tree-based models with hyperparameter optimization (e.g., depth, number of trees, split criteria) and developed a simple prediction workflow/tool that outputs risk estimates from user-provided health indicators.

Clubs/School Activities

Chinese Students and Scholars Association (CSSA), University of Florida Treasurer, Webmaster | 2023–2026

- Managed organization finances as Treasurer by preparing budget requests, tracking reimbursements, maintaining transaction records, supporting nonprofit filing responsibilities, and helping oversee bank account transitions and documentation. Supported Student Government funding and reimbursement processes by reviewing SAR requests, organizing receipts and expense documentation, and monitoring event-related spending to ensure accurate financial records.
- Maintained the organization website and supported event operations as Webmaster by creating PowerPoint materials, Kahoot activities, signup forms, and online resources for member engagement and event coordination.

Awards

Cutler's Award February 2026
Dean's List 2023,2024,2025

Skills

Programming: Python, R, C++, MATLAB, SQL, HTML; RStudio/R Shiny; SAS, SPSS, JMP; Git/GitHub; Excel
Data & Modeling: Statistical modeling, clinical/health data, data cleaning/wrangling, EDA, data visualization, hypothesis testing, ANOVA, experimental design, logistic regression, survival analysis, classification, model evaluation
Languages: English (Native), Mandarin (Native)